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Computational evaluation using machine learning for analysis of membrane desalination process powered by solar energy

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Numerical simulation of mass transfer in membrane contactors for water purification was carried out in this study. The models were developed to estimate permeate volume, a critical performance variable in membrane desalination. We analyzed and compared the accuracy of models: Multilayer Perceptron (MLP), Extreme Gradient Boosting (XGBoost), and a Stacking ensemble method combining MLP and XGBoost. Data preprocessing techniques were applied to improve model performance, including outlier detection using Z-scores and log transformation for normalization. A distinctive contribution of this work is the innovative application of the Jellyfish Optimizer (JO) for hyperparameter tuning, which enhanced model robustness and set it apart from conventional optimization approaches. Unlike conventional optimization algorithms, JO's biologically inspired dynamics offers a novel balance between exploration and exploitation, leading to superior tuning of model hyperparameters. The models' predictive accuracy was evaluated using R^2, RMSE, and MAE with an 85%–15% train–test split. The Stacking model outperformed single models, achieving R^2 values of 0.99060 (training), 0.97627 (cross-validation), and 0.99043 (testing), along with the lowest test errors (RMSE = 6.17921), confirming its efficiency in predicting permeate volume. The results show that stacking significantly improves prediction accuracy in complex tasks like membrane desalination.

Keywords Model, Machine learning, Separation, Removal, Membrane

Integration of membrane water treatment into solar energy has been recognized as a new technology in water purification which can be applied for treatment of water in rural areas[1]. Porous membranes are preferred for application with solar energy due to lower energy consumption compared to nonporous membranes. Electrodialysis (ED)[2] and membrane distillation (MD)[3] are two important membrane processes which can be used in combination with solar energy. ED is an electrical driven membrane system, whereas MD is a temperature driven system for separation of water impurities. In MD for water desalination, thermal energy constitutes the primary energy requirement.

Recently, various types of MD have been developed for water treatment such as direct-contact membrane distillation (DCMD), vacuum membrane distillation (VMD), and air-gap membrane distillation (AGMD)[4–7] with their inherent characteristics. For analysis of MD processes, computational methods have been developed such as computational fluid dynamics (CFD) for finding temperature and concentration distribution in the process. CFD is a powerful tool which can be used for process troubleshooting, design, and optimization[8]. Using this method, governing equations of the process are solved via an appropriate numerical scheme, and the desired variables are obtained in the process. Due to the complexity of CFD, models based on machine learning (ML) have been recently applied for simulation of MD process to analyze mass transfer. These models need some data for training which has been provided by a CFD simulation, so-called hybrid modeling strategy[9,10]. This strategy is more efficient compared to CFD modeling of membranes in terms of solution time and fitting accuracy which make them suitable for process control as well.

Therefore, there is a need for further development of ML algorithms in prediction of MD processes powered by solar energy applicable in water purification to assess its mass transfer performance. This research gap is addressed in this study by utilization of some machine learning models to predict permeate volume in solar-based MD process. Pressure and time were selected as input variables, while permeate volume was predicted based on these parameters.

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Machine learning offers advantages over conventional computational tools by enabling systems to learn from data through algorithms that iteratively improve model accuracy. Some models have been recently developed for energy and environmental applications based on machine learning[11–13]. This field aims to develop algorithms capable of autonomously analyzing experimental data and improving iteratively through model training. Regression analysis plays a crucial role in predicting continuous numerical outcomes from input data, making it a cornerstone of predictive modeling[14]. Unlike CFD-based models, which are grounded in mechanistic principles for simulating processes, ML techniques are data-driven and lack an explicit mechanistic foundation. However, they offer substantial advantages for process optimization, requiring significantly lower computational cost.

In this study, we employed three ML techniques, i.e., Multilayer Perceptron (MLP), Extreme Gradient Boosting (XGBoost), and a Stacking ensemble method combining MLP and XGBoost to predict permeate volume in membrane desalination processes integrated into solar energy. With its deep learning architecture allowing flexibility in capturing intricate patterns within the data, MLP was chosen since it can replicate complicated, non-linear relationships between input features. XGBoost, on the other hand, was chosen for its high scalability and performance in supervised learning tasks, where its sequential tree-building approach enables the model to minimize residual errors iteratively, making it highly effective in regression tasks. The decision to employ Stacking was driven by its potential to improve predictive accuracy by combining the strengths of both MLP and XGBoost. Combining these models and a rigorous hyperparameter tuning process using the Jellyfish Optimizer (JO) was selected to ensure optimal model performance in predicting permeate volume, a highly dynamic and complex process. The novelty of this work lies in employing the JO for hyperparameter tuning during model training. Unlike real-time adjustment approaches, our use of JO focuses on systematically navigating the search space to identify optimal parameters before deployment, ensuring enhanced stability and predictive robustness of the trained models. In addition, we emphasize the advantage of the stacking ensemble strategy: while the MLP effectively captures complex nonlinear relationships, XGBoost excels at identifying dominant structured trends. Their combination through stacking integrates these complementary strengths, enabling more accurate and generalizable predictions than either model alone.

This paper's contributions include exploring the integration of ML models—specifically MLP, XGBoost, and a Stacking ensemble approach—to predict permeate volume, demonstrating the effectiveness of combining deep learning with gradient boosting techniques for enhanced prediction accuracy in membrane process. A key novel aspect of this work is the application of the JO for hyperparameter tuning, which is inspired by the unique movement patterns of jellyfish in ocean currents. This innovative optimization technique improves model performance by efficiently navigating the search space for optimal hyperparameters, further enhancing the predictive capabilities of the models. Additionally, the research investigates the impact of data preprocessing, including outlier detection and log transformation, in refining model inputs to boost the overall accuracy of predictions. The study not only establishes the superiority of the Stacking ensemble model in comparison to individual models but also demonstrates the potential of ML for complex, practical applications in desalination processes. This approach sets a new benchmark for integrating multiple ML methods and optimization techniques in environmental and industrial prediction tasks.

Materials and methods

Membrane data

The process under investigation in this work is a membrane desalination unit which operates using solar energy to produce potable water. Mass transfer across the membrane is a critical parameter, quantified by measuring the permeate volume collected on the permeate side of the membrane unit. This study employs a publicly accessible dataset consisting of 45 observations. This dataset was extracted from[15] which reported modeling and experimental works on measuring permeate volume in solar-based membrane desalination process. Each observation contains two predictors—Time (min) and Pressure (kgf/cm^2)—as well as a response variable, Permeate Volume (mL). The inputs have been selected from the analysis of process and evaluation of their effects and importance on the permeate volume of purified water.

Figure 1 presents violin plots for all variables in the membrane desalination data, providing a visual representation of their distributions, density, and spread. The plot for Time (min) likely illustrates whether the data is symmetrically distributed or skewed, with wider sections indicating a higher density of observations at specific time intervals. Similarly, the Pressure (kgf/cm^2) violin plot may reveal whether the values are concentrated around a specific range or exhibit multiple peaks, suggesting a bimodal distribution. The Permeate Volume (mL) plot is expected to show the distribution of the response variable, highlighting potential outliers and variations. Figure 2 shows histograms illustrating the distributions of all variables.

Outlier detection

Z-score approach was used for detecting outliers to enhance the models' accuracy and robustness. It aids in discovering hidden patterns by pinpointing observations that deviate considerably from the rest[16]. The Z-score method is one of the possible techniques for detecting outliers and applied in this study for the dataset prior to ML modeling.

The Z-score of a given data point X_i in a dataset X is determined using the dataset's standard deviation σ and mean μ, as expressed in the following Eqs[17,18]:

$$Z_i = \frac{X_i - \mu}{\sigma}$$

The Z-score measures the divergence of a data point from the mean in standard deviation units. Data points with Z-scores exceeding 2.5 or 3 are typically regarded as outliers[18]. This method is widely used for identifying

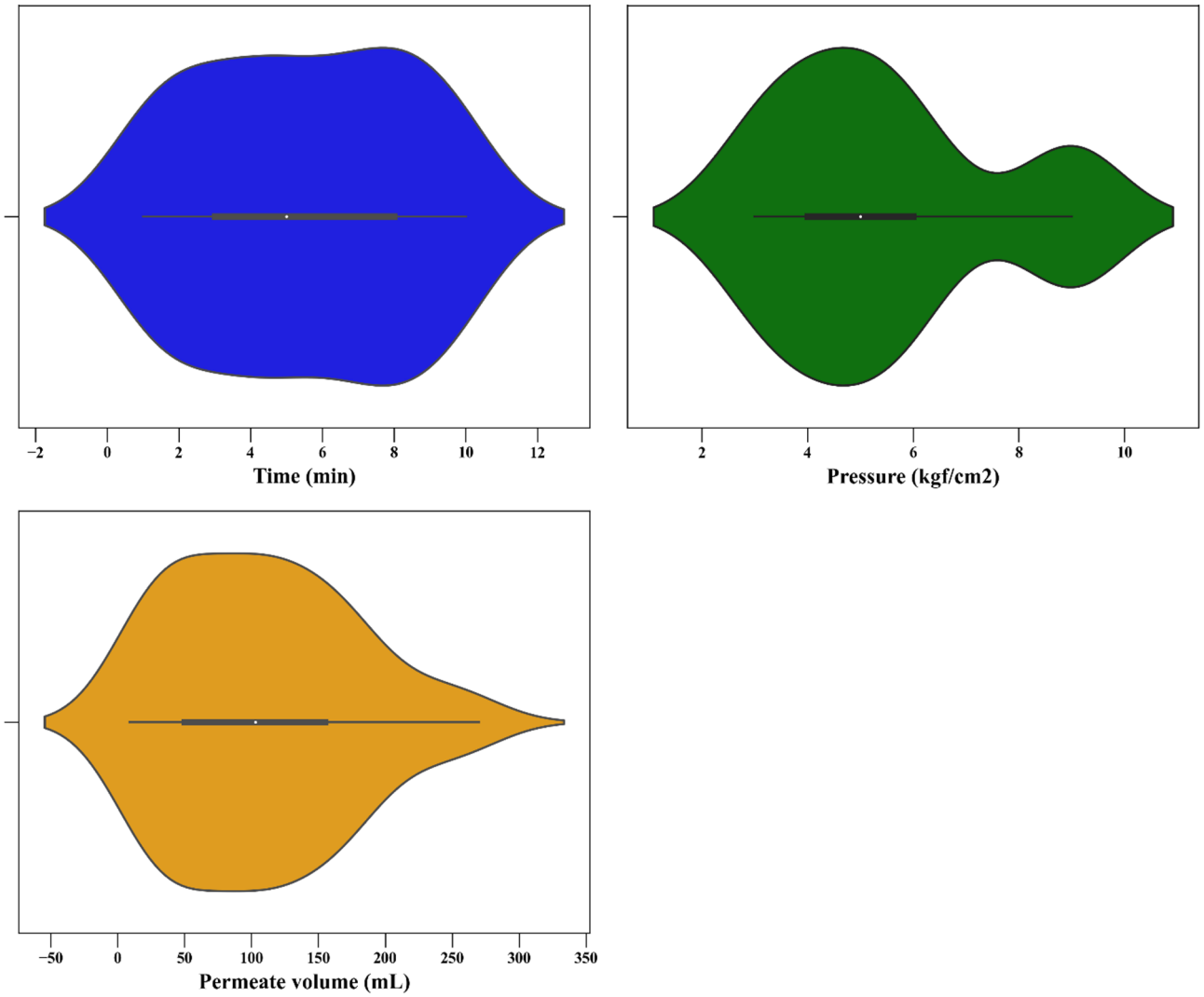


Fig. 1. Violin plots for three membrane desalination variables.

outliers in a dataset. First, we determine the Z-score for all of the data points, which, in terms of standard deviations, indicates how far that point is from the mean. Then, we set a threshold, which is 3 standard deviations in this study. Any data point with a Z-score beyond this threshold is classified as an outlier. This approach offers a definitive and quantitative means to identify anomalous data points, facilitating their removal and enhancing the accuracy and generalizability of models.

Log transformation for normalization

Log transformation is a common technique for normalizing data, particularly when dealing with right-skewed distributions. It helps reduce the effect of large values and brings the data closer to a normal distribution, which is often beneficial for regression models and statistical analysis[19]. The transformation is defined as follows:

$$X' = \log(X + c)$$

where the original variable is denoted by X, X' represents the transformed variable, and c is a small constant added to avoid issues with zero values.

The selection of log transformation for normalization is justified by the observed skewness in the data distributions, as indicated by the violin plots (Fig. 1) and histograms (Fig. 2). The distributions of the variables, particularly the Permeate Volume, are right skewed with a long tail or show significant variability and log transformation helps normalize the data by minimizing the effect of extreme deviations[20].

Multilayer perceptron (MLP)

MLPs are the general term used for feedforward neural networks with several hidden layers. The backpropagation algorithm is a technique utilized for training MLPs and is based on correcting errors during the learning process. This method modifies the model's parameters to reduce the error function, improving performance[21,22]:

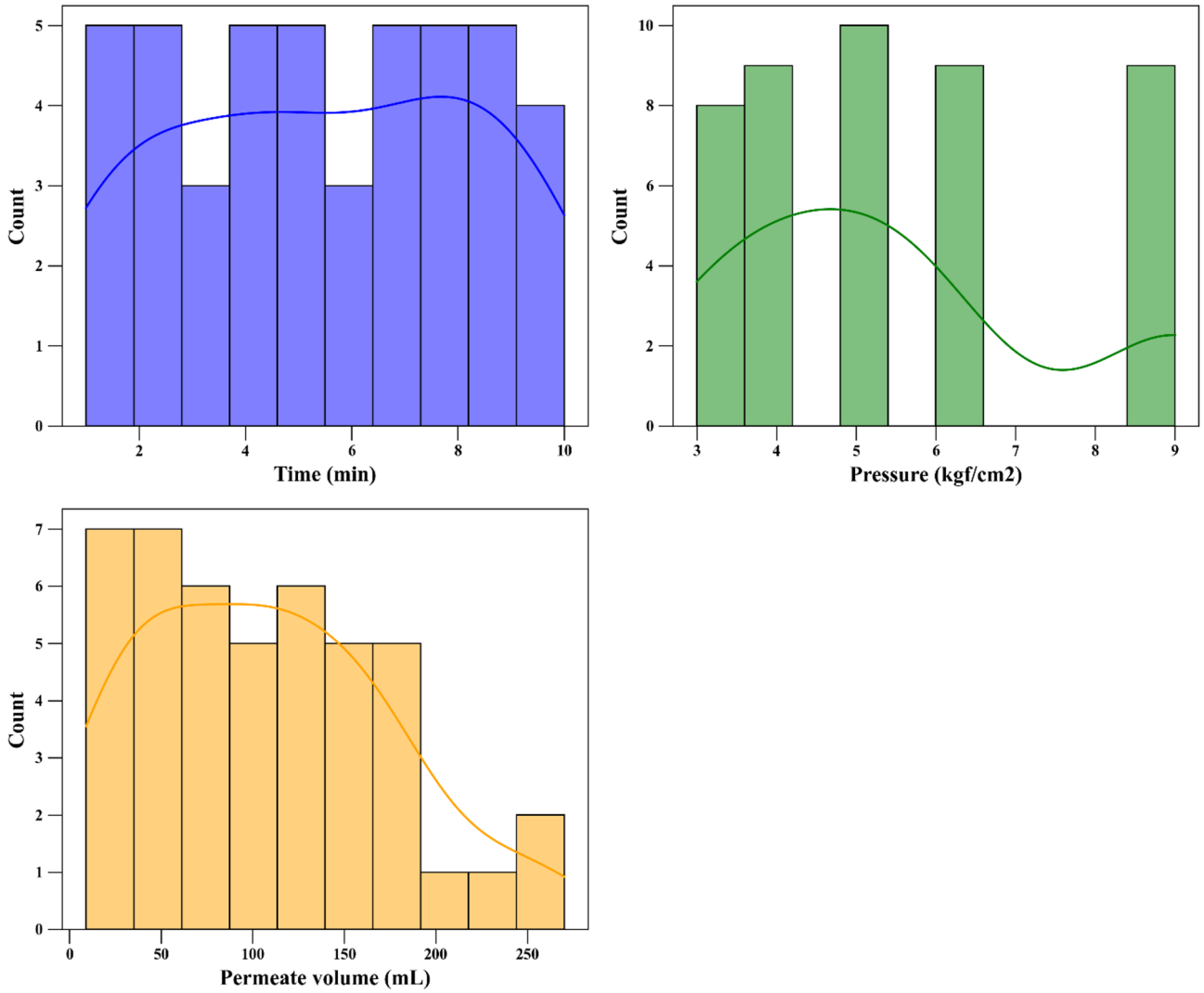


Fig. 2. Histograms of distribution for pressure, time, and permeate volume.

In MLPs, the learning and fitting begins at the input layer and proceeds sequentially through the network, culminating in the output layer where the neurons generate the final result. A layer positioned between the input and output layers is termed a hidden layer. Optimizing these parameters is crucial for achieving optimal performance[22–24].

For an MLP with a single hidden layer and one output, the prediction is given by[22,25]:

$$y = \delta_2 \left(\sum_{i=1}^{m} w_i^{(2)} \delta_1 \left(\sum_{j=1}^{n} x_j w_j^{(1)} + b^{(1)} \right) + b^{(2)} \right)$$

In where y is the predicted vector, m the number of data vectors, and n the number of input features. $w^{(1)} and w^{(2)}$ denote weights between the input–hidden and hidden–output layers, respectively, with biases $b^{(1)}$ and $b^{(2)}$. δ_1 and δ_2 represent the activation functions of the hidden and output layers[22,26].

Extreme gradient boosting (XGBoost)

XGBoost is a modern algorithm, extensively employed for supervised learning tasks. Its key advantages include high scalability, rapid processing, and outstanding performance, making it a preferred option in both ML competitions and real-world applications. The algorithm builds an ensemble of decision trees in sequence, where each tree corrects the errors of the one before it, leading to a more powerful predictive model[27,28]. The core idea behind XGBoost is to optimize a differentiable loss function by iteratively adding trees that minimize the residual errors. The system is trained using the gradient descent method, which calculates the gradients of the loss function based on the current ensemble predictions.

The mathematical formulation of XGBoost can be expressed as follows:

Given a dataset with *n* samples and *m* features, let y_i be the target value and x_i be the feature vector for the *i*-th sample. The model aims to predict $\widehat{y_i}$ employing an ensemble of *K* decision trees:

$$\widehat{y_i} = \sum_{k=1}^{K} f_k\left(x_i\right)$$

where f_k stands for the *k*-th tree in the ensemble. The function to be optimized is:

$$\mathrm{Obj}\left(\theta\right) = \sum_{i=1}^{n} L\left(y_i, \widehat{y_i}\right) + \sum_{k=1}^{K} \Omega\ \left(f_k\right)$$

with $L\left(y_i, \widehat{y_i}\right)$ measuring prediction error and $\Omega\ \left(f_k\right)$ penalizing tree complexity using L1 (Lasso) and L2 (Ridge) regularization.

Stacking ensemble method

Stacking is a technique in ensemble learning that merges several base models to form a superior meta-model. Using the unique strengths of various models enhances overall prediction accuracy. In this research, we applied a stacking approach with XGBoost and MLP as the foundational models[29,30].

The stacking method involves training multiple unique base models utilizing the identical dataset. A meta-model is subsequently developed to produce the final output by employing the predictions from the base models. This strategy enables the meta-model to determine the optimal way to integrate the base models' outputs, which can uncover intricate patterns and enhance overall performance[31]. The stacking ensemble method can be summarized in the following steps:

1. Base Model Development: Train the base models (XGBoost and MLP in this case) on the training dataset. Each base model independently learns to generate predictions.

2. Prediction Generation: The trained base models produce predictions on the validation set, which subsequently serve as input features for the meta-model.

3. Meta-Model Development: The meta-model is trained utilizing the outputs of the base models as input features, while the actual target values serve as the output.

4. Final Prediction: Initially, the trained base models produce predictions on the test dataset, which are subsequently integrated by the meta-model to yield the final output.

Hyperparameter tuning using Jellyfish Optimizer (JO)

Inspired by jellyfish behavior, researchers developed the JO algorithm, where jellyfish movement is influenced by ocean currents[32]. Jellyfish are drawn to ocean currents due to the rich food supply they offer. The direction of the ocean current, denoted as $\overrightarrow{CO}$, can be calculated by taking the mean of the position vectors corresponding to all jellyfish individuals[32,33].

$$\overrightarrow{d} = \frac{1}{n}\sum \overrightarrow{d}_i = \frac{1}{n}\sum \left(X_{best} - e_c X_i\right) = X_{best} - e_c\frac{\sum X_i}{n} = X_{best} - e_c\mu \\ = X_{best} - df$$

where *n* signifies the population size, X_{best} indicates the optimal position attained, μ represents the population positions mean, *df* denotes the distance[33]. As a result, the updated position can be calculated as[32,34]:

$$X_i\left(t+1\right) = r\left(X_{best} - \beta\ .r\right)\mu\ \ + \ X_i\left(t\right)$$

In this instance, β is a distribution coefficient exceeding 0 and r denotes a random value within the interval (0, 1)[33].

JO is an ideal choice because of its ability to balance exploitation and exploration, drawing inspiration from the movement patterns of jellyfish in ocean currents. By navigating complex search spaces effortlessly, this feature allows JO to efficiently identify the optimal hyperparameters that improve model performance (see Fig. 3)[34].

Results and discussion

The models' predictive performance was assessed using an 85%–15% train-test split. The training set, consisting of 85% of the dataset, was utilized to train the MLP, XGBoost, and Stacking models. Performance metrics, including MAE, RMSE, and R^2 score, were determined for both the training and test sets to compare model accuracy. Additionally, cross-validation was performed to ensure robustness and mitigate overfitting. The results highlight the predictive capabilities of each model and demonstrate the efficacy of ensemble learning in improving prediction accuracy. Although the dataset comprises only 45 observations, its adequacy is justified by the low-dimensional nature of the input space (two predictors: Time and Pressure) and the high consistency of model performance across cross-validation, training, and test sets. The close alignment of R^2 scores and error metrics across these splits (e.g., stacking model R^2: 0.99060 training, 0.97627 cross-validation, 0.99043 test) indicates that the models generalize well without overfitting. Nevertheless, the small dataset size remains a limitation, and future work should validate the models on larger and more diverse datasets to confirm their robustness in broader contexts.

Table 1 presents the assessment metrics of the final models—MLP, XGBoost, and the Stacking model—based on training, cross-validation, and test data. The results demonstrate that the stacking model consistently outperforms both MLP and XGBoost across all evaluation criteria. Specifically, the stacking model achieves the highest R^2 (0.99060 on training, 0.97627 in cross-validation, and 0.99043 on test data), demonstrating superior predictive accuracy. Additionally, the stacking model reports the lowest MAE and RMSE values, indicating improved error minimization compared to the individual models. Given these findings, the stacking approach

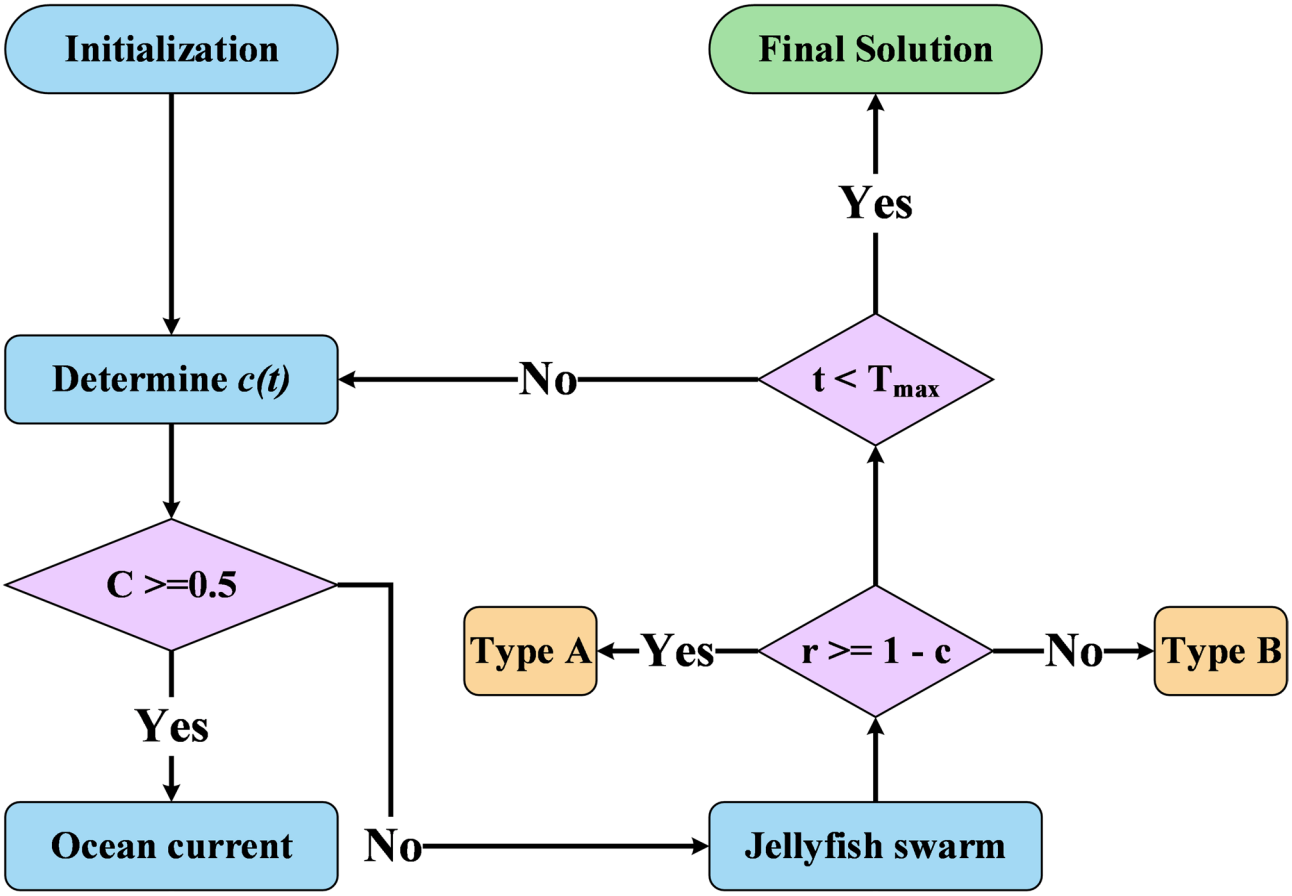


Fig. 3. Workflow of the jellyfish optimizer (JO)[33].

Metric		MLP model	XGBoost model	Stacking model
Train	R^2 score	0.96340	0.95768	0.99060
	RMSE	11.9418	12.8410	6.05188
	MAE	9.00181	10.1286	4.17026
Cross validation	R^2 score	0.95062	0.87272	0.97627
Test	R^2 score	0.96242	0.90211	0.99043
	RMSE	12.2469	19.7669	6.17921
	MAE	9.60500	14.6429	5.90072

Table 1. Statistical assessment of all models for estimation of membrane permeate volume.

effectively integrates the strengths of MLP and XGBoost, making it the most appropriate model for accurate prediction.

The absence of overfitting in the stacking regression model is evident from its cross-validation R^2 score of 0.97627, which remains close to its training ($R^2 = 0.99060$) and test ($R^2 = 0.99043$) scores. This consistency across different datasets confirms the model's accuracy. Moreover, the superiority of the stacking model is visually substantiated by Figs. 4, 5 and 6 which revealed expected and predicted values for MLP, XGBoost, and the stacking model. The stacking ensemble exhibits the strongest alignment between predicted and observed outcomes, indicating enhanced predictive precision and robustness relative to each of the standalone models.

Figure 7 presents the residual plot for the stacking regression model, reinforcing its status as the best-performing model in estimation of membrane permeate volume. Residual plots are essential for assessing model accuracy, as they display the differences between expected and predicted values. In this case, the residuals for both the training and test datasets are symmetrically distributed around zero, without significant patterns or heteroscedasticity. The absence of large residuals and their even spread further confirms that the model maintains high accuracy, validating its superiority over the MLP and XGBoost models.

Although the stacking model exhibits strong generalization with no signs of overfitting—as evidenced by the close alignment of R^2 scores across training (0.99060), cross-validation (0.97627), and testing (0.99043)—some

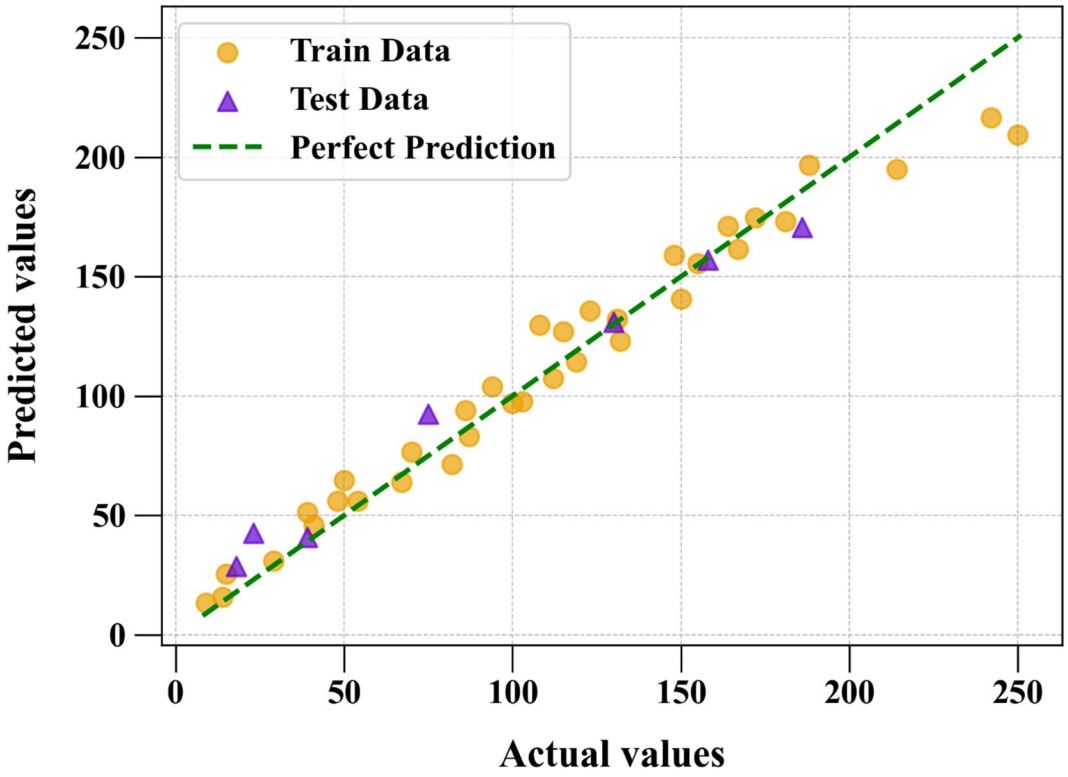


Fig. 4. Comparison between experimental and predicted values of permeate volume (MLP model).

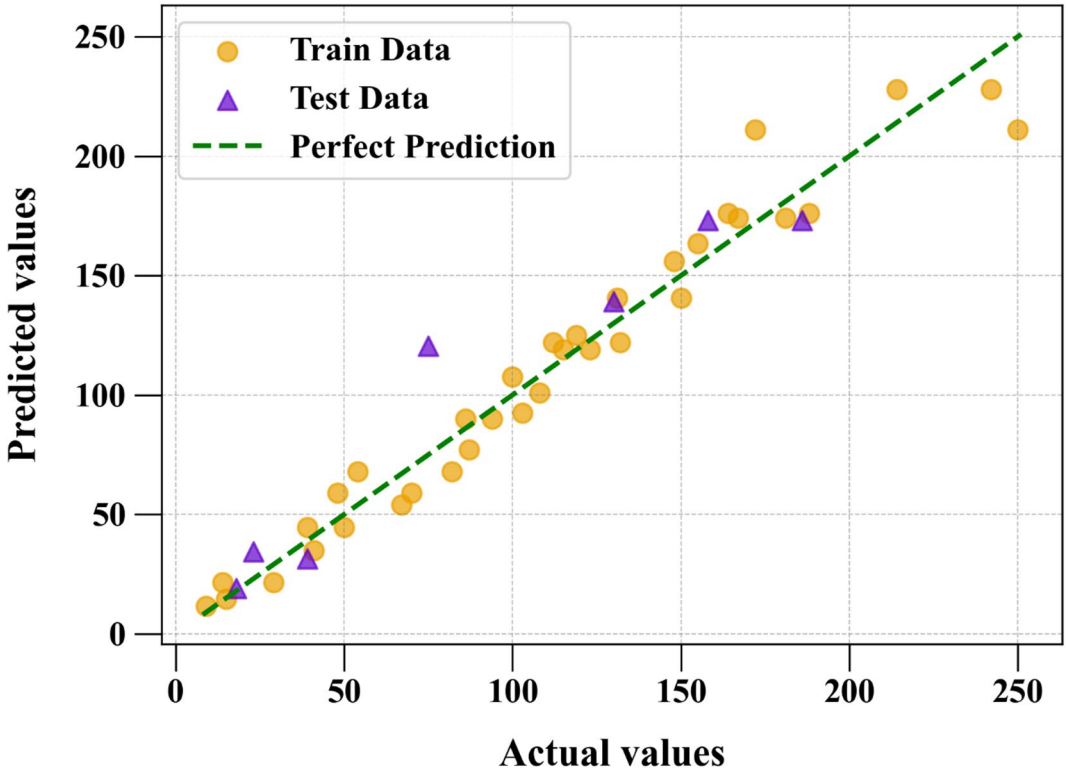


Fig. 5. Comparison between experimental and predicted permeate volumes (XGBoost model).

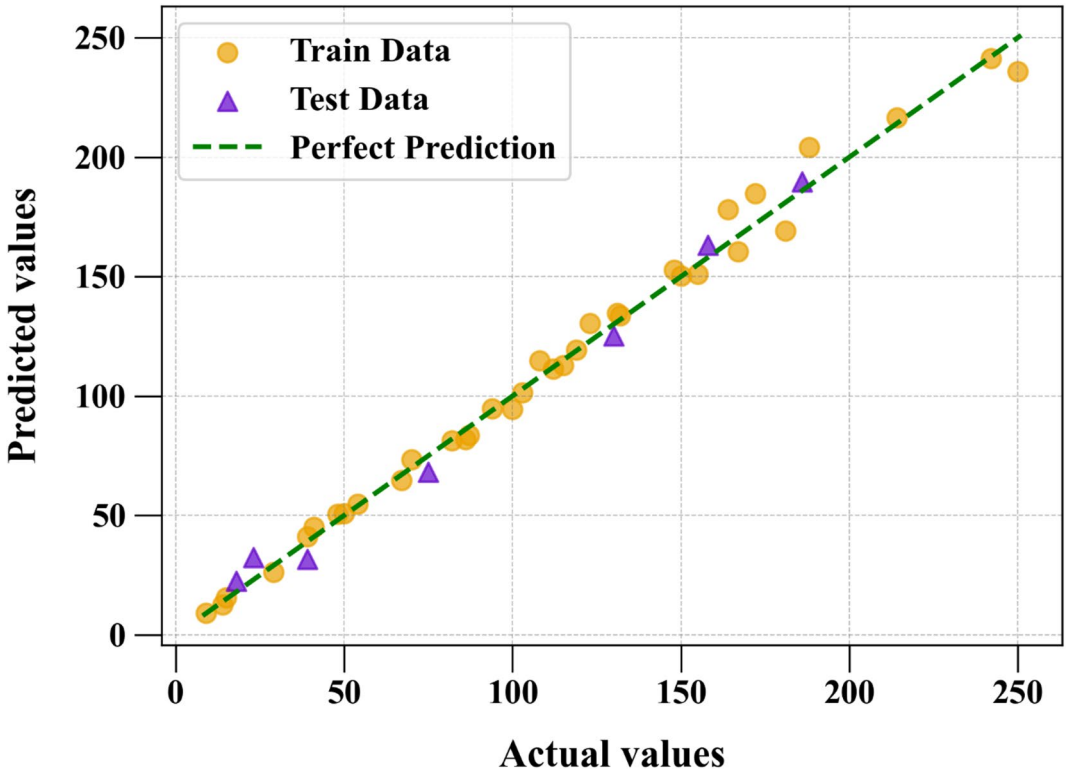


Fig. 6. Comparison between experimental and predicted permeate volumes (stacking model).

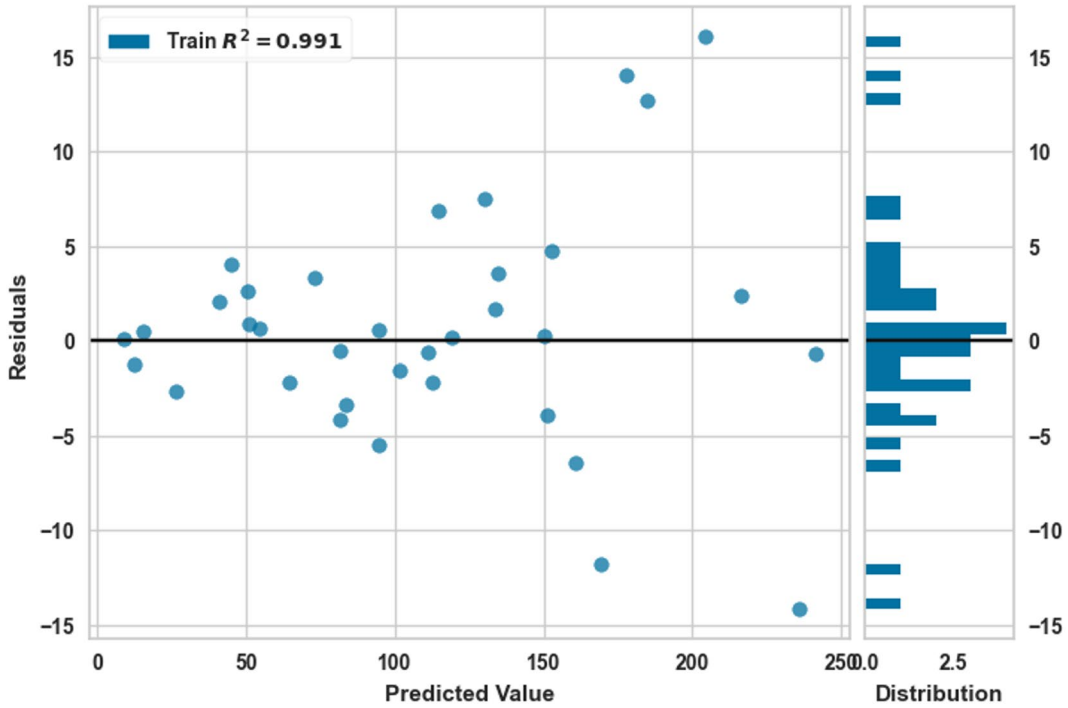


Fig. 7. Residual plot of stacking regression model (both train and test data).

prediction variability may arise from factors not captured in the model, such as membrane fouling dynamics, temperature variations, or feed water quality. While the residual plots show a well-balanced error distribution, the relatively small dataset (45 observations) may limit the model's exposure to a wider range of operating conditions.

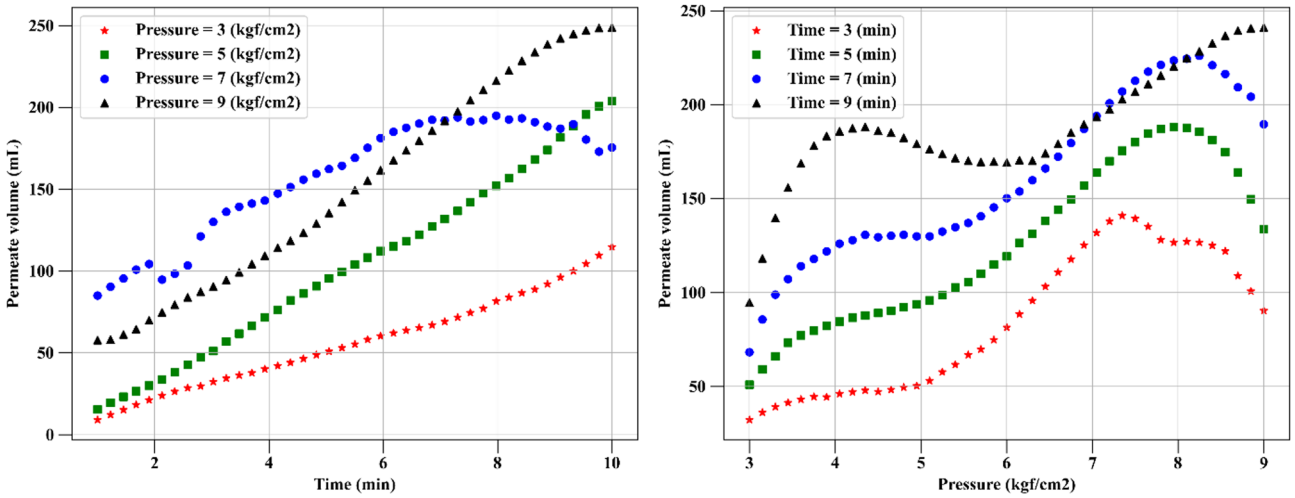


Fig. 8. Individual impacts of time and pressure on the predicted membrane permeate volume.

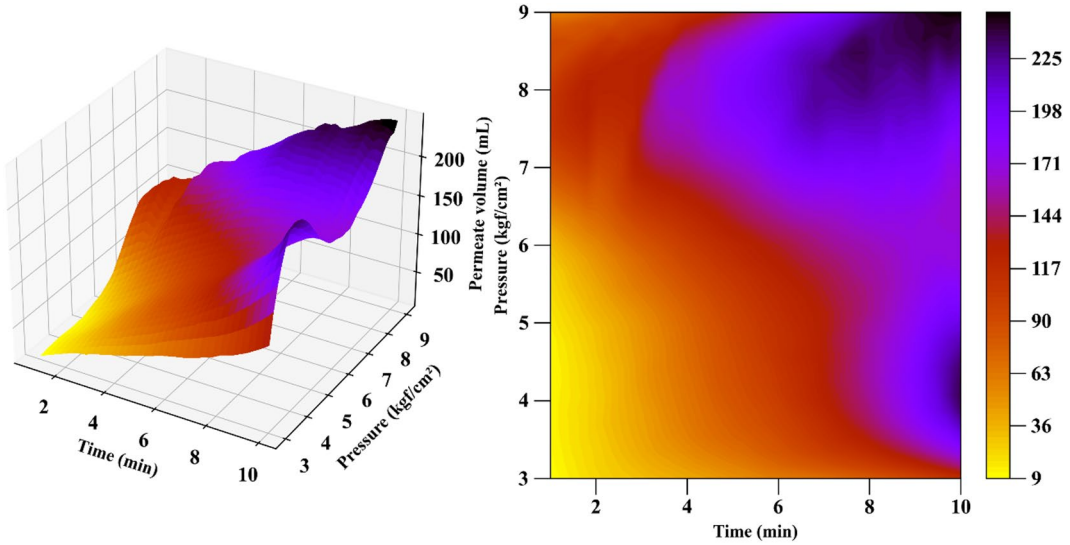


Fig. 9. Two-dimensional and three-dimensional representations of the predicted permeate volume as a function of time and pressure.

The superior performance of the Stacking model is related to the complementary strengths of its base learners—MLP and XGBoost. MLP, with its deep learning architecture, excels at capturing non-linear relationships and subtle patterns within the data, particularly when interactions between input features are complex. XGBoost, on the other hand, is highly effective at handling structured data and identifying dominant trends through its gradient-boosted decision trees. By combining these two models, the stacking ensemble leverages MLP's capacity for function approximation and XGBoost's robustness and interpretability. The meta-model in the stacking architecture learns to weigh their predictions in a way that mitigates individual weaknesses, leading to reduced bias and variance and ultimately improved generalization performance.

Figure 8 highlights the individual impacts of Time and Pressure on the predicted output, showing almost smooth and consistent trends. Also, Fig. 9 provides 2D and 3D visualizations of the predicted permeate volume as a function of Time and Pressure. As seen, both inputs show direct relationship with the membrane permeate volume, so that the highest volume of permeate is obtained at the highest time and pressure. To achieve optimal membrane operation, a cost–benefit analysis is necessary, as increasing pressure improves permeate production but also raises energy consumption. Therefore, the optimal operating point should be determined by evaluating both total energy consumption and associated costs. It is also seen that the slope of variations becomes less sharp at longer time (> 9 min) which could be due to the membrane fouling by the organic or inorganic materials in the feed water which caused concentration polarization. The fouling of membrane reduces mass transfer rate through the pores and significantly decreases the separation efficiency.

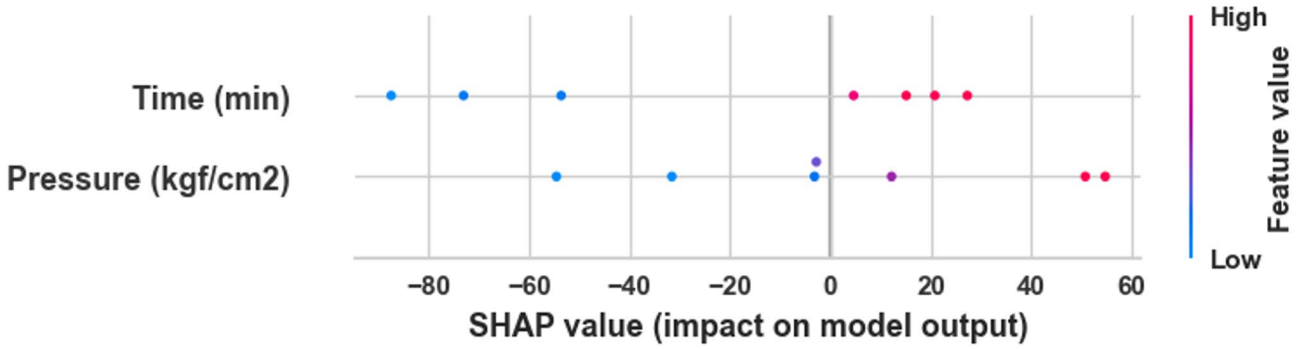


Fig. 10. SHAP analysis results for time and pressure features.

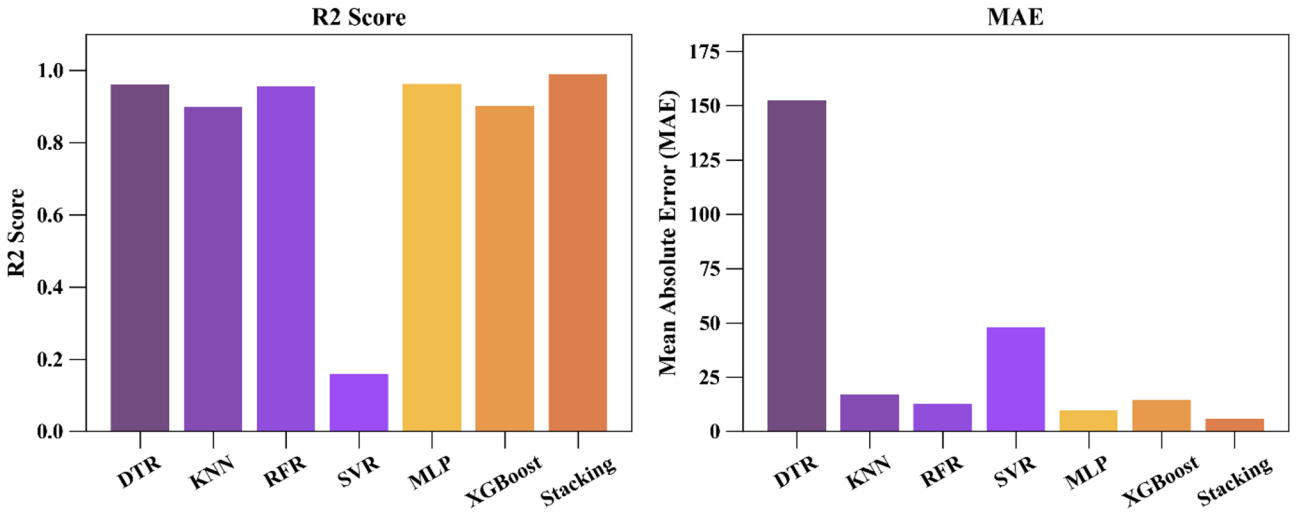


Fig. 11. Comparison of the models in this study with existing models from[15].

Figure 10 presents the SHAP (SHapley Additive exPlanations) summary plot for two inputs, which interprets the contribution of each input feature—Time and Pressure—to the model's predictions of membrane permeate volume. The plot quantifies the impact of these features on the output, ranking them by importance and showing the direction and magnitude of their influence. As illustrated, Time has the highest contribution to the prediction variability, followed closely by Pressure. The color gradient indicates the feature value (from low to high), and its position along the x-axis reflects whether it increases or decreases the predicted permeate volume of membrane desalination unit. This visualization confirms that both inputs are influential, but Time exerts a more pronounced effect on permeate volume, supporting findings from Figs. 8 and 9.

Figure 11 compares the performances of the models in this study with the baseline models from[15] which developed some ML models in prediction of permeate volume. The stacking model outperforms individual models like MLP and XGBoost in terms of R^2 score and MAE across training, cross-validation, and test data. Specifically, it achieves the highest R^2 (0.99060 in training, 0.97627 in cross-validation, and 0.99043 in testing), indicating better accuracy and generalization. The stacking method integrates the strengths of MLP and XGBoost, leading to more robust predictions and minimizing errors when evaluated against the other models. This demonstrates the superior performance of stacking in this study, particularly in comparison to the baseline models in[15].

The findings of this study confirm that combining ensemble learning with bio-inspired optimization can deliver highly accurate predictions for complex desalination processes. The Stacking ensemble consistently outperformed individual models by integrating the complementary strengths of MLP and XGBoost, while the JO provided a distinctive contribution through its ability to efficiently tune hyperparameters during training. These results establish the novelty of this work in both methodological design and application. However, certain Limitations must be acknowledged. The dataset comprised only 45 observations, which restricts the diversity of operating conditions available for training. Although cross-validation, preprocessing, and ensemble modeling helped mitigate overfitting and demonstrated robust generalization, the small sample size remains a critical limitation in applying machine learning to real-world scenarios. Additionally, the study focused exclusively on pressure and time as input variables, while other important factors such as temperature fluctuations, fouling

effects, and feedwater quality were not explicitly modeled. Addressing these constraints with larger datasets and additional features will be essential for further validating and extending the approach.

Conclusion

This study presented a machine learning framework for predicting permeate volume in solar-powered membrane desalination processes. Among the evaluated models, the Stacking ensemble combining MLP and XGBoost consistently outperformed individual models, achieving R^2 values above 0.99 and the lowest error metrics. A key novelty of this work is the integration of the Jellyfish Optimizer (JO) for hyperparameter tuning during model training, which significantly improved predictive robustness compared to conventional optimization methods. The ensemble approach further enhanced performance by utilizing the complementary strengths of MLP (capturing nonlinearities) and XGBoost (capturing structured trends), demonstrating the value of combining diverse models for complex process prediction.

Despite these advances, the study is constrained by the Limited dataset of only 45 observations. While the issue of overfitting was mitigated through preprocessing, cross-validation, and ensemble learning, the small sample size remains a critical limitation for machine learning applications. Future work should validate the approach with larger and more diverse datasets, explore additional ensemble strategies, and extend the application of JO-based tuning to other environmental and industrial prediction tasks.

Data availability

The dataset and generated codes used during the current study are freely available from the link: https://github.com/TareqNafeaAlharby/perm_vol_prediction.

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Declarations

Competing interests
The authors declare no competing interests.

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